MULTIVARIABLE CALCULUS • UNIT 8.01

C Surface Integrals, Divergence Theorem & Stokes Theorem

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 B	Q4 C	Q5 C
Q6 D	Q7 B	Q8 B	Q9 C	Q10 D
Q11 B	Q12 A	Q13 A	Q14 A	Q15 D
Q16 B	Q17 A	Q18 A	Q19 D	Q20 D
Q21 D	Q22 B	Q23 B	Q24 D	Q25 C

Surface Integrals, Divergence Theorem & Stokes Theorem Practice Exercises

1 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ 1,\ 0)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

- **A** e^2 ✓
- **B** $1 + e^2$
- **C** $-e^2$
- **D** $2e^2$

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2yz + ye^x$; at $(2,\ 1,\ 0)$ this is e^2.

2 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ -2,\ 1)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

- **A** -4
- **B** 2 ✓
- **C** 3
- **D** -2

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2z$; at $(2,\ -2,\ 1)$ this is 2.

3 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ -2,\ 0)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

- **A** $1 - 2e^2$
- **B** $-2e^2$ ✓
- **C** $2e^2$
- **D** $-4e^2$

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2yz + ye^x$; at $(2,\ -2,\ 0)$ this is $-2e^2$.

4 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ 1,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

A -3

B 4

C -4 ✓

D -8

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2z$; at $(2,\ 1,\ -2)$ this is -4.

5 MCQ

Find $\nabla \times \mathbf{F}$ at $(2,\ 0,\ 1)$ for $\mathbf{F} = \langle y, -x, z^2 \rangle$.

A $\langle 0, 0, 2 \rangle$

B $\langle 0, -2, 0 \rangle$

C $\langle 0, 0, -2 \rangle$ ✓

D $\langle 1, 1, -1 \rangle$

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, -2 \rangle$; at $(2,\ 0,\ 1)$ this is $\langle 0, 0, -2 \rangle$.

6 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(-2,\ 1,\ -2)$ for $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$.

A 9

B -5

C 6

D -6 ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2x + 2y + 2z$; at $(-2,\ 1,\ -2)$ this is -6.

7 MCQ

Find $\nabla \times \mathbf{F}$ at $(-2,\ -2,\ 2)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

A $\langle -6, 0, -2 + e^{-2} \rangle$

B $\langle 6, 0, 2 - e^{-2} \rangle$ ✓

C $\langle 0, 2 - e^{-2}, 6 \rangle$

D $\langle 7, 1, 3 - e^{-2} \rangle$

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle -x + z^2, 0, z - e^x \rangle$; at $(-2,\ -2,\ 2)$ this is $\langle 6, 0, 2 - e^{-2} \rangle$.

8 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(0,\ -2,\ 0)$ for $\mathbf{F} = \langle y, -x, z^2 \rangle$.

A -2

B 0 ✓

C 1

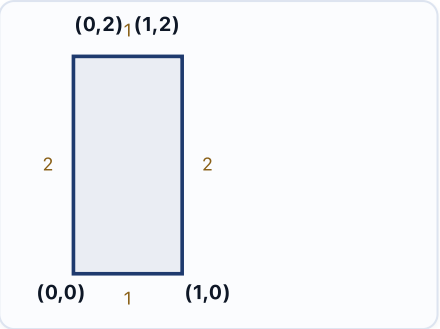
D 2

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2z$; at $(0,\ -2,\ 0)$ this is 0.

9 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.



A $\sqrt{14}$

B $3\sqrt{14}$

C $2\sqrt{14}$ ✓

D $4\sqrt{14}$

WORKED SOLUTION & VERIFICATION

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the surface area is $2\sqrt{14}$.

10 MCQ

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V \nabla \cdot \mathbf{F}\, dV$.

A 1

B 2

C 4

D 3 ✓

WORKED SOLUTION & VERIFICATION

The vector field is $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. The divergence of $\mathbf{F}$ is $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$. The volume V enclosed by the unit cube is given by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$. The volume of this cube is $1 \times 1 \times 1 = 1$. Therefore, the flux is given by the triple integral: $\iiint_V \nabla \cdot \mathbf{F}\, dV = \iiint_V 3\, dV$. Since the integrand is a constant, this is equal to $3 \times \text{Volume}(V) = 3 \times 1 = 3$.

11 MCQ

Find $\nabla \times \mathbf{F}$ at $(0, -2, -1)$ for $\mathbf{F} = \langle x^2 y, -xy^2, z \rangle$.

A $\langle 0, 0, 4 \rangle$

B $\langle 0, 0, -4 \rangle$ ✓

C $\langle 0, -4, 0 \rangle$

D $\langle 1, 1, -3 \rangle$

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, -x^2 - y^2 \rangle$; at $(0, -2, -1)$ this is $\langle 0, 0, -4 \rangle$.

12 MCQ

Find $\nabla \times \mathbf{F}$ at $(0,\ -2,\ 2)$ for $\mathbf{F} = \langle xy, yz, xz \rangle$.

- **A** $\langle 2, -2, 0 \rangle$ ✓
- **B** $\langle -2, 0, 2 \rangle$
- **C** $\langle 3, -1, 1 \rangle$
- **D** $\langle -2, 2, 0 \rangle$

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle -y, -z, -x \rangle$; at $(0,\ -2,\ 2)$ this is $\langle 2, -2, 0 \rangle$.

13 MCQ

Calculate $\nabla \cdot \mathbf{F}$ at $(2,\ 2,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

- **A** -4 ✓
- **B** -3
- **C** 4
- **D** -8

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2z$; at $(2,\ 2,\ -2)$ this is -4.

14 MCQ

Compute $\nabla \cdot \mathbf{F}$ at $(1,\ 1,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

- **A** -4 ✓
- **B** -3
- **C** 4
- **D** -8

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2z$; at $(1,\ 1,\ -2)$ this is -4.

15 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(-2,\ 0,\ 0)$ for $\mathbf{F} = \langle yz, xz, xy \rangle$.

- **A** 1
- **B** 2
- **C** -1
- **D** 0 ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 0$; at $(-2,\ 0,\ 0)$ this is 0.

16 MCQ

Calculate $\nabla \cdot \mathbf{F}$ at $(0,\ -2,\ -1)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

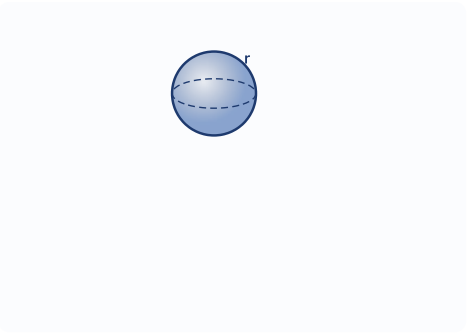
- **A** -4
- **B** 2 ✓
- **C** 3
- **D** -2

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 2yz + ye^x$; at $(0,\ -2,\ -1)$ this is 2.

17 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.



- **A** $\frac{2916\pi}{5}$ ✓
- **B** $\frac{5832\pi}{5}$
- **C** $\frac{1458\pi}{5}$
- **D** $-\frac{2916\pi}{5}$

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = \frac{2916\pi}{5}$.

18 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, 0 \rangle$ around the boundary of the square $[0, 2] \times [0, 2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

- **A** 8 ✓
- **B** 16
- **C** -8
- **D** 4

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 8$.

19 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle 2x, -y, 3z \rangle$ across the sphere $x^2 + y^2 + z^2 = 1$.

- **A** $\frac{8\pi}{3}$
- **B** $-\frac{16\pi}{3}$
- **C** $\frac{32\pi}{3}$
- **D** $\frac{16\pi}{3}$ ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 4$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = \frac{16\pi}{3}$.

20 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, z \rangle$ around the boundary of the square $[0, 1] \times [0, 1]$ in the plane $z = 0$, oriented counterclockwise seen from above.

- **A** -2
- **B** 1
- **C** 4
- **D** 2 ✓

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 2$.

21 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x+y, y+z, x+z \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.

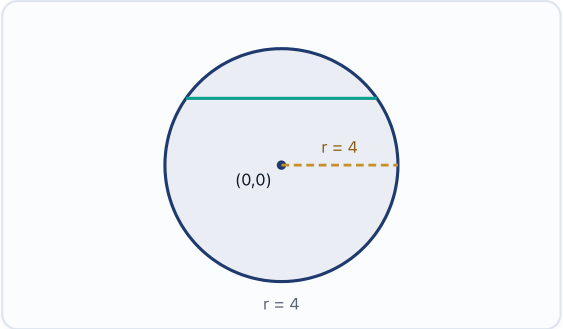
- A -108π
- B 216π
- C 54π
- D 108π ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 108\pi$.

22 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -2y, 2x, y \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.



- A 8π
- B 4π ✓
- C -4π
- D 2π

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 1, 0, 4 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 4\pi$.

23 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

- A 64π
- B 32π ✓
- C 16π
- D -32π

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 32\pi$.

24 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the closed cylinder $x^2 + y^2 \leq 1,\ 0 \leq z \leq 1$.

A $\frac{3\pi}{2}$

B -3π

C 6π

D 3π ✓

WORKED SOLUTION & VERIFICATION

$\nabla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E \nabla \cdot \mathbf{F}\, dV = 3\pi$.

25 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -3y, x, z \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.

A 2π

B 8π

C 4π ✓

D -4π

WORKED SOLUTION & VERIFICATION

$\nabla \times \mathbf{F} = \langle 0, 0, 4 \rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dS = 4\pi$.